

Photoabsorption and Photoionization in Gases

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Abstract: A compact demonstration paper for testing the paper compilation pathway. We summarize the distinction between photon absorption and photoionization in a dilute gas and introduce the simplest attenuation model used to connect microscopic cross sections with macroscopic interaction probabilities.

Keywords: photoabsorption; photoionization; gas detectors; cross sections.

Introduction

Photoabsorption denotes any process in which an incident photon transfers its energy to the medium, whereas photoionization is the subset of such interactions that leaves a free electron and a positive ion. In gaseous detectors the distinction matters because absorption controls photon transport, while ionization can seed additional charge and therefore modify the detector response.

Minimal interaction model

For a uniform gas with number density n , path length L , and an energy-dependent microscopic cross section $\sigma(E)$, a simple independent-interaction model gives

$$P(E, L) = 1 - \exp(-n\sigma(E)L).$$

Using $\sigma_{\text{abs}(E)}$ gives the total absorption probability, while $\sigma_{\text{ion}(E)}$ describes only the ionizing branch. Consequently, $\sigma_{\text{ion}(E)} \leq \sigma_{\text{abs}(E)}$ whenever non-ionizing absorption channels are available.

Photon regime	σ_{abs}	σ_{ion}
Below threshold	finite or small	0
Near threshold	increasing	turn-on
Above threshold	channel dependent	finite

Table 1: Illustrative qualitative behaviour; entries are intentionally schematic.

Photon transport context

In practical gaseous systems the relevant attenuation length depends strongly on photon energy, mixture composition, pressure, and the availability of ionizing and non-ionizing channels. The distinction becomes particularly useful when photons generated in an avalanche can either disappear harmlessly or create a new electron that participates in subsequent multiplication.

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Discussion

The simple exponential model above neglects scattering, secondary emission, excited-state kinetics, and spatial variations of the gas. Nevertheless, it provides a useful baseline for checking units, threshold behaviour, and the relation between microscopic cross sections and detector-scale probabilities. A realistic implementation can replace the schematic cross sections by tabulated or calculated data without changing the structure of the transport model.

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Scope

This document is deliberately synthetic and exists to exercise the shared paper infrastructure, metadata parsing, style switching, two-column Elsevier layout, and output routing. Its scientific content can later be replaced by a full treatment without changing the compilation workflow.

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